HACK@BVP 8.0 · HEALTH AND WELL BEING · PS6: THE STOCKOUT SHIELD

SwasthyaGrid

A district medicine engine that forecasts demand at every facility, flags stock-out and expiry risk early, and proposes the transfer or reorder a named officer approves

TEAM IDEAFORGE

Team member	Institution	Role
Harsh Kawatra	Delhi Technological University	Team Leader
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Watch the product walkthrough ↗

Our answer: move before you buy. Stock-outs and expiry share one cause: nobody sees a district's medicine as one stock. SwasthyaGrid forecasts every shelf and moves surplus and near-expiry stock to where it will be used, before anyone orders more.

1.36 crore stock-out days avoided a year, base estimate

72.2% → 86.1% key-medicine availability

1 Problem: shelves run empty while state stores write off expired stock

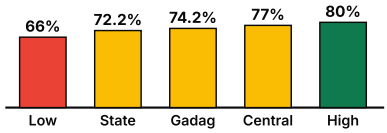
66 to 80% of key medicines in stock at public facilities studied

₹390.26 Cr of drugs expired in Telangana's stores, 2016-22

₹168 Cr of drugs expired in Odisha's stores, 2019-25

43.4% of health spending paid out of pocket, often at private chemists

KEY-MEDICINE AVAILABILITY, BY STUDY



No study finds availability above 80%. Missing antivenom matters: India records 58,000 snakebite deaths a year.

Root cause: stock is recorded, but never seen or moved district-wide

- **Seen late:** HMIS reports monthly, after the shelf empties
- **No forecast:** burn rates, seasons and surges go uncounted
- **Not moved:** surplus at a nearby PHC goes unseen
- **Not rotated:** near-expiry stock never reaches busier PHCs
- **Hiring is slow:** 79.9% of CHC specialist posts sit empty

2 Solution: one district view that forecasts, flags and proposes, and a named officer decides

1 **Sense** — Stock, receipts and issues from existing state exports

→ 2 **Forecast** — P50 and P90 demand for each facility and medicine

→ 3 **Flag** — Stock-out risk in the resupply window and units set to expire

→ 4 **Propose** — Sized transfer, reorder or rotation, with reasons

→ 5 **Approve** — A named officer approves, edits or rejects every move

→ 6 **Learn** — Receipt is confirmed; outcomes sharpen the next forecast

Officer approves every move · Stock counts only, no patient data · State data stays with the state · Hindi and English voice for PHC staff

Precedent: eVIN cut vaccine stock-outs 80% once stock became visible in 733 districts.

3 Approach: each PS6 requirement mapped to a method and an output

PS6 requirement	Method	What the officer sees
R1 · Demand forecast per facility	Consumption rebuilt from the stock ledger (opening + receipts − closing). TSB intermittent-demand method for sparse items; a LightGBM quantile model on recent use, season and calendar for the rest.	P50 and P90 demand; days of cover as a range, e.g. 4.2 days (2.9 at P90)
R2 · Stock-out and expiry risk	Chance that P90 demand over the resupply lead time exceeds stock on hand; a first-expiry-first-out batch projection of units still on the shelf at each expiry date.	A risk band, plus units and ₹ value at risk of expiry
R3 · Seasons, outbreaks, sudden change	Monsoon, heat and festival calendar features; IDSP weekly syndromic counts and IMD rain and heat as surge inputs; a CUSUM change-point test on recent consumption.	The surge raises the forecast itself, and the trigger is named
R4 · Transfers and reorder quantities	Min-cost-flow over the district road network (OSRM) that minimises stock-out days, value expired and transport cost. Donors keep 14 days of cover; cold-chain items travel cold; reorders follow an order-up-to rule at P90 lead-time demand.	Ranked moves, each with quantity, source, km and minutes
R5 · Sparse, irregular, incomplete data	Pooling across similar facilities and medicines for thin histories; freshness flags on stale entries; outlier clipping; gaps filled from stock deltas, never invented.	Confidence that drops when data is thin or stale
R6 · Explain every recommendation	The engine writes reason codes (cover, P90 demand, expiry date, donor surplus, distance) and a stated confidence score. Language models only phrase them in Hindi or English; they never compute a number.	One plain-language reason line on every proposal

Stack: Python and FastAPI · LightGBM and statsmodels · OR-Tools · OSRM road routes · Next.js dashboard · Firestore · Sarvam voice (saaras STT, Sarvam 105B LLM, bulbul TTS) for Hindi, Hinglish and English · Gemini 3.5 Flash Lite for typed answers

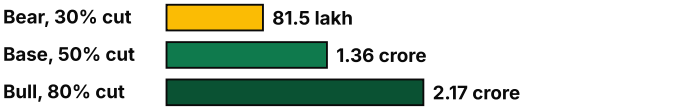
4 Expected impact: an estimated 1.36 crore stock-out days avoided a year

1.36 crore stock-out days avoided a year across 38,241 PHCs and CHCs, base case (half of eVIN's 80% cut)

72.2% → 86.1% key-medicine availability

17,412 stock-out days avoided per district a year

₹558 Cr expired in two audited states; rotation targets it



SDG 3 · medicine access · SDG 9 · resilient supply · SDG 12 · less waste

5 Evaluation: aimed at the organisers' data against a burn-rate baseline

Test	How we measure it, and the target
Forecast accuracy	Rolling-origin backtest WAPE per facility and medicine; beat the burn-rate baseline on every medicine group
Calibration	Weeks with demand at or below the P90 forecast; close to 90%
Stock-outs and expiry	Replay held-out months with and without the engine's moves; fewer stock-out days and less ₹ expired than baseline
Explanations	Proposals with a reason and a confidence score; 100%
New shock	Organiser-injected surge or missing-data block; the forecast adjusts within one review cycle

Sources: availability, PMC8654139 and PMC10292169 · expiry, CAG Telangana (Aug 2024) and CAG Odisha (Oct 2026) · out-of-pocket share, NHA 2022-23 · snakebite, eLife 2020 · facilities and specialist posts, MoHFW HDI 2022-23 · eVIN, UNDP India. **Impact model:** 38,241 facilities × 7 tracer medicines × 365 days × 27.8% unavailability × a 30%, 50% or 80% cut (80% is eVIN's result); about 780 districts. Estimates, not measured results.